

Avance — Presenter Kit

Everything a teammate who didn't build this needs to present it: the pitch in plain words, a demo script, what every technical term actually means, and answers to the questions judges are likely to ask.

Team: Avance

Lead: K N Anirudh

avance-tufk.vercel.app

One-liner	Avance turns one big assignment into small trackable steps, so students always know what's next and teachers see who's falling behind days before the deadline — not after.
Live demo	avance-tufk.vercel.app
Demo logins	Password for every account: <code>password123</code> Student view → <code>student1@avance.dev</code> (up to <code>student12@avance.dev</code>) Teacher / host view → <code>host@avance.dev</code> Org head view → <code>head@avance.dev</code>
Demo org	CSE-A (join code <code>CSEA-2025</code>) — project: <i>"Build a C Banking Management System"</i>
Team	K N Anirudh (Lead), Daivaansh Chandok, Akash Sharma, Divya Shree N, Keshav, Vedant

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01 • The problem

Today, a teacher hands out an assignment, and then **doesn't see anything** until the final file lands in their inbox. If a student is stuck, confused, or hasn't started, nobody finds out until it's too late to help.

Progress happens invisibly, between the assignment and the deadline.

Say this in your own words: *"By the time a teacher finds out a student is struggling, it's usually the deadline itself that told them — and by then it's too late to actually help."*

02 • The idea

Avance breaks one assignment into a short sequence of **checkpoints** — for example: Requirements → Design → Implementation → Testing → Final Submission. Every student in a group gets the exact same checkpoints for the exact same project, which is what makes group progress comparable at all.

That single change gives both sides something they didn't have before:

- **Students** always know exactly what to do right now, how much is left, and where they stand versus their group.
- **Teachers** get a live view of the whole class or group: who's on track, who needs attention, and — importantly — *which step* of the assignment is tripping up the most people.

FRAMING LINE FOR THE PITCH

"Not a to-do list with XP — a progress-visibility system with a motivation layer." XP, streaks and the leaderboard exist to keep students moving through *real* checkpoints; they are never the point of the product by themselves.

03 • Who uses it

Student	Joins a group, sees their current project as a linear checkpoint path, submits evidence (a PDF) per checkpoint, and tracks their own XP, streak and rank.
Group Host (lecturer)	Creates a project, either builds checkpoints manually or lets the AI Quest Builder propose them, sets deadlines, reviews submissions, and watches the Command Center.
Organisation Head	Manages the organisation itself (e.g. a whole department) and decides who is allowed to act as a Host.

04 • Demo script

This is the flow the pitch deck itself is built around. Practice it end to end at least once before presenting — it takes about 4–5 minutes unhurried.

1. **Open on the landing page, signed out.** Read the one-liner off the screen: "A game layer for real-world work." Don't linger — this is context-setting, not the demo.
2. **Log in as a student** (`student1@avance.dev`). Land on the dashboard and point at the "Current Quest" card: project title, a progress bar, the current checkpoint, a deadline countdown, and their rank. Say: "This is the one screen a student needs — the next action is never a mystery."
3. **Open the project** to show the linear checkpoint path: completed steps, the current one, and future steps locked/greyed out. If there's a pending submission, show the PDF upload flow.
4. **Log out, log back in as the host** (`host@avance.dev`). Go to **Organisations** → **CSE-A** → **Command Center**. This is the single most important screen in the demo.
5. **Point at the traffic-light counts** (On Track / At Risk / Critical) and then the per-checkpoint completion chart. Land on: "Testing is sitting at 43% — that tells the lecturer exactly where the class is getting stuck, three days before the deadline, not three days after."
6. **Show the AI Quest Builder** (creating/editing a project): paste or upload an assignment, show the proposed checkpoint list clearly labelled as AI-generated and fully editable, then click **Approve & Publish**. Say out loud that nothing reaches students without this human approval step.
7. **Show the group Leaderboard** and a student's **Profile** (XP, streak, rank) to close the loop on the motivation layer.
8. **Optional close:** the personal **Habit Tracker**, framed explicitly as "outside the classroom, doesn't touch grades or institutional data at all" — it's there to show the product's range, not to be the headline.

05 • Feature glossary — in plain words

Checkpoint

One small, clearly defined step inside a bigger assignment — e.g. "Design" or "Testing." Finish every checkpoint in order and the assignment is done.

AI Quest Builder

The tool a teacher uses to turn a pasted or uploaded assignment into a first-draft list of checkpoints automatically — which the teacher then edits and has to approve before anyone sees it.

Command Center

The teacher's one-screen dashboard: how the whole class or group is doing, right now.

On Track ● on track / **At Risk** ● at risk / **Critical** ● critical

A traffic-light status per student or group, calculated only from checkpoints completed versus time remaining — never from anything outside the platform.

Bottleneck analytics

A per-checkpoint completion chart that shows exactly which single step of the assignment is slowing the whole class down.

XP

Points earned for completing checkpoints. Specific to one organisation — your XP in one class doesn't carry over into another.

Streak

How many days in a row a student has made real, checkpoint-based progress — not just opened the app.

Leaderboard

A ranking within your own group, based on checkpoints completed — built to motivate, not to publicly shame whoever's behind.

Fair Play

An optional accountability flag only a teacher can set, always logged, and kept clearly separate from ordinary late or rejected work.

Habit Tracker

A personal, ungraded feature — reading, exercise, coding practice — that never touches organisation data, XP or rank.

06 • Tech stack, explained simply

Nobody expects a presenter to code live — but here's what to say if a judge asks "what's this built with," in words that don't require a CS degree to follow.

Supabase

Our cloud "filing cabinet + front door." It gives the app two things off the shelf: a real online database that stores every organisation, project and checkpoint (so the data lives on the internet, not one laptop), and a secure sign-up/login system, so we didn't write our own password-handling code and risk getting it wrong. It's also why the live demo can run on the internet at all — a simple on-laptop database would lose its data between requests once deployed; Supabase's database doesn't.

Prisma

The translator between our app's code and the Supabase database — it turns normal code into safe database queries and defines what an "Organisation" or "Checkpoint" looks like, in one readable place.

Next.js 16 / React

The framework the whole app is built in. It lets one project serve both what you see on screen and the secure logic behind it — every action that changes real data (submitting work, approving a checkpoint) is double-checked on the server, so nothing can be faked from the browser.

Tailwind CSS v4

The styling toolkit behind the clean, consistent look — small reusable style "utilities" instead of writing custom styling from scratch for every screen.

GSAP

An animation library used for the small, deliberate motion touches — a progress bar filling smoothly, a hover response — that make the app feel considered rather than static.

Three.js / WebGL

The 3D-graphics browser technology behind the soft animated purple background pattern shown after sign-in.

Zod

A safety-check library — it makes sure any data (like what the AI Quest Builder proposes, or what a form submits) is shaped correctly before the app trusts and saves it.

Vercel

The hosting service the live demo actually runs on.

07 • Design & UI/UX choices, explained

The interface is deliberately split into two layers, and that split is itself the design decision worth explaining if asked:

- **The information layer** — dashboards, progress bars, the Command Center — stays clean, restrained and analytical, because that's what earns a teacher's or an institution's trust. No confetti, no giant game buttons, no fantasy-RPG skin.
- **The motion layer** — the soft animated purple background, small hover/click responses, generated avatar art — is where the "game" personality lives. It only appears after sign-in, and it's kept at low opacity specifically so it never competes with or obscures real information.

Other choices worth being able to explain:

- **Linear checkpoints, not a skill tree.** A skill tree is more "game-like" but makes "what do I do next?" harder to answer — for both the student and the teacher watching a whole class. Avance deliberately keeps progression a straight line.
- **Generated avatars, not photo uploads.** Every student gets a fun, randomly generated icon instead of needing to upload a real photo — lower friction, and a bit of personality without an identity/privacy trade-off.
- **Responsive and accessible.** The layout adapts across phone, tablet and desktop; animation respects a user's "reduce motion" setting; every action works through normal keyboard-focusable controls, not effects that require a mouse.

08 • What's actually built, vs. what's roadmap

Say this with confidence — everything below the line is a real, working feature, not a slide-only claim:

- Sign-up / login and organisation / group roles
- AI Quest Builder — accepts PDF, DOCX or pasted text
- Linear checkpoints with locked / current / done states
- PDF submission, with either automatic or host-approval verification
- Command Center — risk status and per-checkpoint bottleneck analytics
- XP, streaks, and a group leaderboard
- A personal habit tracker
- In-app notifications and an audit log for sensitive actions

KNOWN, HONEST LIMITATIONS – ONLY BRING THESE UP IF ASKED DIRECTLY

The AI Quest Builder doesn't call an external AI model. It's a rule-based text analyser we built ourselves, wired behind the same interface a real language model would use later. This was deliberate: the product still has to work perfectly even if an external AI API is slow, down, or rate-limited, so that fallback exists either way — we just haven't wired up the live-model half yet.

Submission files currently live on the app server's disk, not in dedicated cloud storage. Moving them into Supabase's storage — alongside the database we already migrated there — is the next infrastructure step.

09 • Judge Q&A

Answers are written the way you'd actually say them out loud — short, plain, and honest about what's built versus what's a stated direction.

About the idea

Q What problem does Avance actually solve, in one sentence?

A Teachers currently only find out a student is struggling when the final assignment is late or weak — Avance makes the steps in between visible, so problems show up days before the deadline instead of on it.

Q Isn't this just a to-do list with points bolted on?

A A to-do list only helps the one person using it. Avance's real value is on the teacher's side — because every student in a group gets identical checkpoints, a teacher can see at a glance which step the whole class is stuck on, and which specific students need help, before the deadline.

Q How is this different from Google Classroom or Moodle?

A Classroom and Moodle track the start and the end of an assignment — handing it out, then collecting it. They don't show what's happening in between. Avance sits exactly in that gap.

Q How is this different from Duolingo or other gamified apps?

A Those apps gamify a personal habit, but no institution ever sees the result. Avance puts the same kind of motivation (XP, streaks, a leaderboard) on top of real institutional work, and the institution actually sees the progress data that comes out of it.

Q Why break work into checkpoints instead of just grading the final file?

A Grading only the final file means every problem is found on the very last day, when it's too late to fix. Checkpoints turn one big invisible assignment into a handful of small visible stages, so a teacher can see exactly where the class stands, days before the deadline.

About the "AI" in it

Q What AI actually powers the Quest Builder — is it ChatGPT?

A Honestly: for this build it's a rule-based text analyser we wrote ourselves. It reads the assignment for patterns — bullet points, words like "must" or "required," headings — and assembles those into a proposed checkpoint list. It isn't calling an external AI model right now. It's built behind the same interface a real model would plug into later.

Q Isn't that a weakness — "fake AI"?

A It's a reliability decision, not a shortcut: the product has to keep working even if an AI API is ever down or rate-limited, so this fallback would exist either way. And every proposal — AI-powered or not — still needs a human teacher to review and approve it before students ever see it.

Q What stops it from generating something wrong or inappropriate?

A Two safeguards: it only ever proposes, never auto-publishes — the teacher sees an editable preview and must click Approve & Publish; and it's built from a small set of predictable rules, not free-form generation, so at worst it misses a detail rather than inventing one.

About the technology

Q What is Supabase, in plain terms — why did you use it?

A Think of it as a ready-made cloud filing cabinet with a front-door lock. It gives us a real online database — so the app's data lives on the internet, not one laptop — plus a secure login system, so we didn't build our own password handling and risk getting it wrong. It's also why the live demo can run on the internet at all.

Q Is this a real working app, or a clickable mockup?

A It's a real full-stack application — every screen is backed by an actual database and actual server logic. Happy to sign up a brand-new account live and watch a real progress bar get created in front of you.

Q Where do uploaded PDFs go — is that secure?

A Every submission is stored where it can't be reached directly — downloading one always goes through a check that confirms you're actually allowed to see that specific file first.

Trust, privacy & fairness

Q How do you flag a student "At Risk" without spying on them?

A The system never looks outside the platform — no keystrokes, no screen activity, no webcam. It only compares two numbers: checkpoints actually completed versus how much of the deadline's time has passed. It never claims to know why someone's behind, only that progress may need attention.

Q Can a student fake their own XP, rank, or verification status?

A No — every one of those values is calculated and written on the server, never trusted from the browser, and the server independently re-checks who's asking and whether they're allowed to before anything is saved.

Q Does the leaderboard humiliate students who are behind?

A It's built not to — the main comparison is within your own small group doing the same assignment, and the design principle behind it is explicitly "competition without humiliation," not a public ranking of who's worst.

Q What is "Fair Play" and who controls it?

A An optional accountability flag only a teacher can set — never a student, never the AI — and every change to it is logged. It's kept clearly separate from normal situations like a late or once-rejected submission.

Design & UI/UX

Q Why does the app look the way it does — the animation, the purple background?

A We split the interface into two layers on purpose: dashboards and progress data stay clean and professional, because that's what earns a teacher's trust; the animated background and small hover effects are the "personality" layer, and they're kept subtle and only shown after sign-in so they never compete with real information.

Q Why linear checkpoints instead of a skill tree?

A A skill tree is more game-like but makes "what do I do next?" harder to answer, for both the student and a teacher scanning a whole class. Avance deliberately keeps it a straight line.

Scale & viability

Q Would this work for 500 students, not just the 60 in your demo?

A The mechanism doesn't change with size — the Command Center runs on simple aggregate counts, which scale the same way at 60 or 6,000 students. It's actually more useful at larger scale, since that's exactly when a teacher can no longer check on every student by hand.

Q What's the business model — how would a college actually adopt this?

A *(This is a suggested direction, not a built feature — say so if pressed.)* The natural model mirrors tools colleges already pay for, like Turnitin or Moodle Workplace: an institutional subscription priced by active students, not a charge to individual students.

Q What's next on the roadmap?

A Move file storage into Supabase's cloud storage; connect the Quest Builder to a real language model behind its existing interface, keeping the human-approval step unchanged; add direct import from Google Classroom or Moodle.

Is there research behind this?

Q Did your team just guess at what would work, or is there research behind it?

A Three concrete anchors: India's National Education Policy 2020 calls for flexible, learner-centred, technology-enabled education, which is the direction this is built in; the early-warning idea comes from learning-analytics research on tracking students during the work, not just at the end; and the streak mechanics are deliberately modelled on behaviour-change research — the "don't break the chain" and identity-based-habit ideas from James Clear's *Atomic Habits* — because small, visible, immediate progress is what actually keeps people going.

10 • If you get stuck

It's fine not to know an implementation detail — presenters who didn't build the app aren't expected to. Better answers than guessing:

- "That's on our roadmap, not something we've built yet — I don't want to overclaim it on stage." (pairs with Section 08 above)
- "That specific piece was built by a teammate — happy to follow up with the exact detail afterward."
- Never invent a number. If asked for one you don't have, point to Section 12 below, or offer the follow-up line above.

11 • Research grounding & positioning

- **NEP 2020** — India's National Education Policy calls for flexible, learner-centred, technology-enabled education; that's the direction Avance is built in.
 - **Learning analytics / early-warning systems** — research on tracking students *during* the work (formative assessment), not only at the final hand-in.
 - **Gamification-in-education research** — on the engagement effects of streaks and leaderboards, used here as a motivation layer, never a replacement for real work.
 - **Atomic Habits (James Clear)** — the streaks, small linear checkpoints, and always-visible progress are deliberately modelled on identity-based habits and "don't break the chain."
 - **Where Avance sits:** Classroom/Moodle track only the two endpoints of an assignment; Duolingo-style apps gamify personal habits with no institutional visibility. Avance is in the gap between them.
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12 • Numbers to quote

What	Number
Demo class size	60 students
On Track / At Risk / Critical	● 41 on track ● 13 at risk ● 6 critical
Checkpoint completion — Requirements	96%
Checkpoint completion — Design	91%
Checkpoint completion — Implementation	78%
Checkpoint completion — Testing (the bottleneck)	43% △
Checkpoint completion — Documentation	12%
Example student progress	60% through "Build the Banking System," rank #12 of 58
Early-warning framing	flagged Day 11 of 14 → 3 days to intervene, vs. discovering it on Day 14

Prepared for the Avance team's SIH26207 presenters. Source: project README, PRD and codebase, and the pitch deck (Avance_SIH26207_Pitch_v2.pptx). Everything in Section 08's "known limitations" box is deliberately included so no presenter gets caught off guard — it's fine, even good, to be upfront about it if asked.